

Publication package: SiN nanobeam cavity tuned to the Rb D2 line

Final trimmed design: $f_{\text{res}} = 384.2306$ THz (+0.0 GHz from Rb D2), $Q = 11990$, $\kappa/2\pi = 32.0$ GHz, $\beta = 0.886$, $C = 24.4$, $\eta = 0.842$, $V = 0.874$ $(\lambda/n)^3$.

Acceptance gate: PASSED (tolerance $|df| \leq 8.0$ GHz).

Scope: EM-derived scalar proxies (g , C , η , β) from classical FDTD; not a cavity-QED simulation. No gate/readout/entanglement-fidelity claims. See *limitations.md*.

Five-design comparison (one convention)

design	f_res_THz	detuning_GHz	eta_atom	beta_wg	C_atom	Q
design_1	385.2411	1010.5	0.083	0.087	21.2	1137.0
invDes(design_1)	383.9390	-291.6	0.278	0.287	45.0	1778.0
seed	383.6151	-615.5	0.088	0.089	471071.2	27116903.9
invDes(seed)	384.5341	303.5	0.834	0.875	25.4	12600.4
final	384.2306	0.0	0.842	0.886	24.4	11990.1

Fig. 1

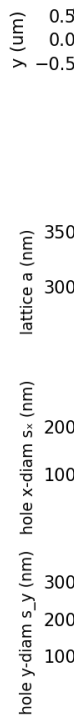


Fig. 1 - Device geometry and per-hole parameter schedule (seed vs final).

Fig. 2

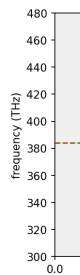


Fig. 2 - Bloch band structure of the mirror and defect cells; band gap, light line, Rb D2 target, and final f_{res} .

Fig. 3



Fig. 3 - Resonance fit and the waveguide-extraction loss budget (κ_{wg} dominant; residual = $1 - \beta$).

Fig. 4



Fig. 4 - Mode localization and field at the atom position.

Fig. 5



Fig. 5 - Tuning sweeps (global / defect / output) and the overcoupling Pareto map.

Fig. 6

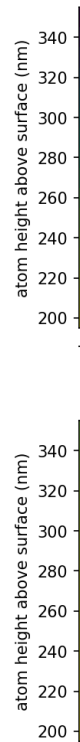


Fig. 6 - Atom-position robustness of eta and C.

Fig. 7

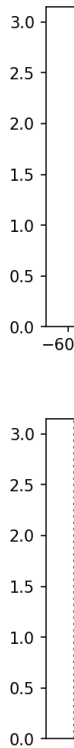


Fig. 7 - Fabrication-tolerance robustness (corner grid), unretuned and retuned.

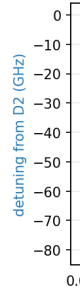
Fig. 8



Fig. 8 - Five-design comparison in one metric convention.

Supplement A

Supplement A - Numerical convergence (mesh / PML / run-time).



Supplement B



Supplement B - Mode isolation.